

DESIGN AND ANALYSIS OF ALGORITHMS

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Experiment 17: Selection Sort

Code:

```
def selection_sort_min_writes(arr):
    arr = arr.copy()
    swaps = 0
    n = len(arr)
    for i in range(n):
        min_index = i
        for j in range(i + 1, n):
            if arr[j] < arr[min_index]:
                min_index = j
        if min_index != i:
            arr[i], arr[min_index] = arr[min_index], arr[i]
            swaps += 1
    return arr, swaps

data = [23.5, 19.2, 25.1, 18.8, 21.4]
sorted_data, swap_count = selection_sort_min_writes(data)
print("Sorted Data:", sorted_data)
print("Number of Swaps:", swap_count)
```

Output:

```
=== RESTART: C:/Users/hk838/Downloads/Slot-
Sorted Data: [18.8, 19.2, 21.4, 23.5, 25.1]
Number of Swaps: 2
```